SE LAB – 1

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1 .Problem Context and Overview :

Hostel administration requires a geo-tagged ticketing pipeline to report plumbing, electrical, and carpentry issues, dispatch maintenance staff, track SLA timelines, and record resident closure sign-offs.

Target stakeholders / actors given: Hostel Resident, Maintenance Warden.

2 . Requirements Table :

Exactly five Functional Requirements and two Nonfunctional Requirements.

REQ_ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE(SHORT)
FR-001	Functional	The system shall allow residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass:Ticket is created with a unique ID and routed to the warden queue. Fail: Incomplete form without room tag is accepted.	Core maintenance reporting
FR-002	Functional	The system shall allow the maintenance warden to assign reported tickets to maintenance staff.	High	Pass: Selected staff member is assigned and the ticket displays the assigned staff. Fail: Ticket remains unassigned after successful assignment.	Enables maintenance dispatch
FR-003	Functional	The system shall allow residents to provide a closure sign-off after their maintenance		Pass: Resident can see statuses such as Open,	Keeps residents informed

		issue is resoved.	Medium	Assigned, In Progress, and Resolved. Fail: Status is not displayed for an existing ticket.	
FR-004	Functional	The system shall allow the maintenance warden to update the status of maintenance tickets.	High	Pass: Updated status is saved and visible to the resident. Fail: Status change is not recorded.	Tracks maintenance progress
FR-005	Functional	The system shall allow residents to provide a closure sign-off after their maintenance issue is resolved.	Medium	Pass: Resident can confirm closure and the ticket is marked as closed. Fail: Ticket cannot be closed after resident confirmation.	Confirms issue resolution
NFR-001	Non-Functional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without status change	High	Pass: Alert is triggered after 24 hours without status change. Fail: No alert is generated after the SLA is exceeded.	Prevents delayed maintenance
NFR-002	Non-Functional	The system shall protect resident and ticket information from unauthorized access.	High	Pass: Unauthorized users cannot access restricted ticket information. Fail: Restricted information is accessible without authorization.	Ensures data security

3 . Peer Critique And Revision :

Clarity: Is the “shall” statement unambiguous?

Testability: Does the acceptance criterion give a clear PASS/FAIL result?

Relevance: Does the requirement align with the hostel maintenance scenario?

Comments: Add a short suggestion if clarification is needed.

Swap back and revise the table after peer feedback.

4 . Actors and Use Cases :

The lab requires at least three actors and at least five use cases.

- Actor 1 – Hostel Resident: Login, Raise Maintenance Ticket, View Ticket Status, Provide Closure Sign-off.
- Actor 2 – Maintenance Warden: Login, View/Manage Tickets, Assign Maintenance Staff, Update Ticket Status, Track SLA.
- Actor 3 – Maintenance Staff: Work on assigned maintenance issues and update ticket status. This actor is derived from the scenario's maintenance-staff dispatch function to satisfy the lab's three-actor requirement.

Suggested use cases:

- UC-01 Login
- UC-02 Raise Maintenance Ticket
- UC-03 View Ticket Status
- UC-04 Provide Closure Sign-off
- UC-05 View / Manage Tickets
- UC-06 Assign Maintenance Staff
- UC-07 Update Ticket Status
- UC-08 Track SLA
- UC-09 Trigger Escalation Alert
- UC-10 Add Room Number / Geo-tag
- UC-11 Attach Photo
- UC-12 Categorize Issue Severity

6. Include / Extend Explanation

«include» = the included behavior is always required as part of the main use case.

Examples:

- Raise Maintenance Ticket → «include» Add Room Number / Geo-tag
- Raise Maintenance Ticket → «include» Attach Photo
- Raise Maintenance Ticket → «include» Categorize Issue Severity

«extend» = optional or conditional behavior.

Example: Trigger Escalation Alert → «extend» Track SLA, because escalation occurs only when the high-priority 24-hour SLA condition is met.

MAIN SUCCESS SCENARIO:

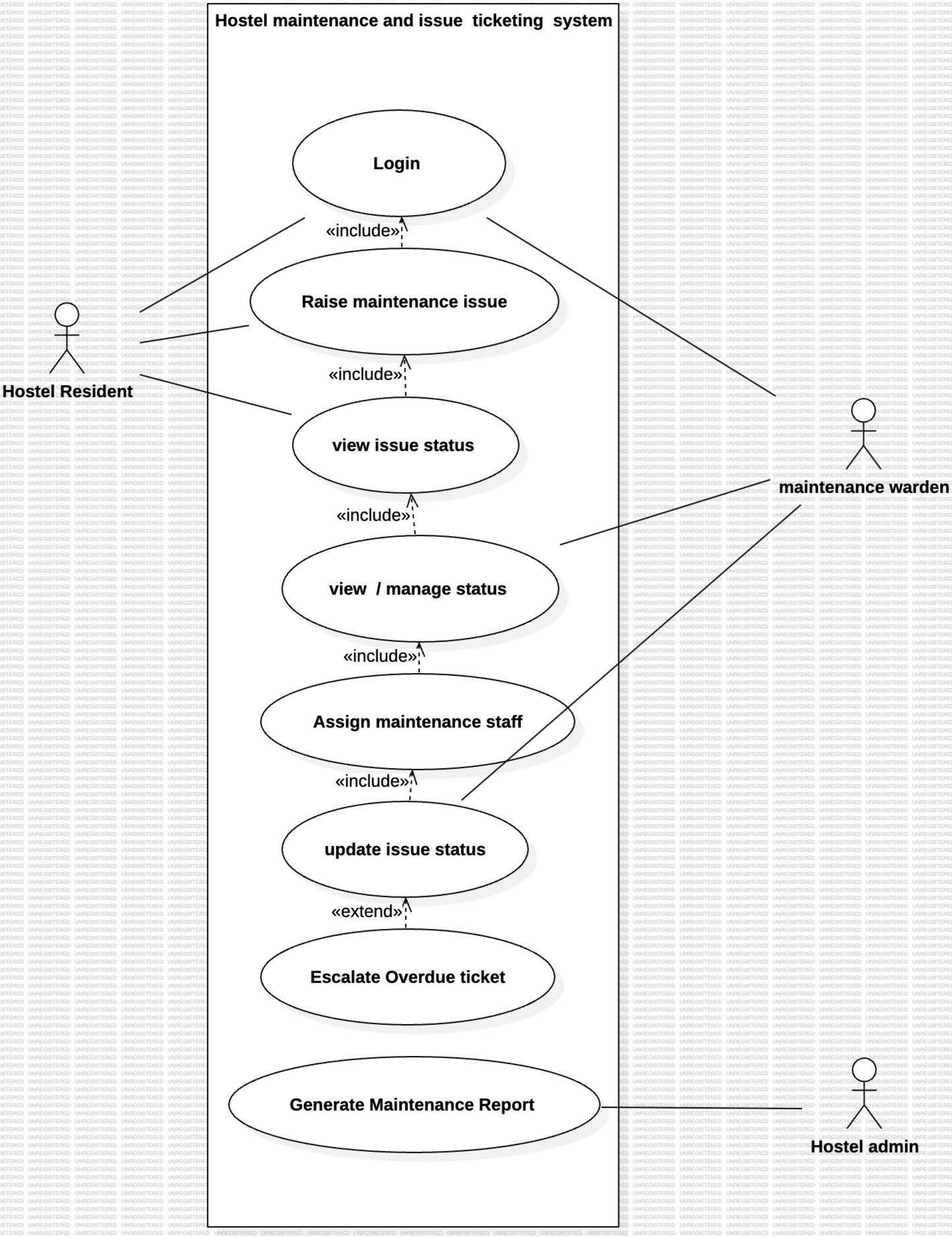
1. Resident logs in.
2. Resident selects “Raise Maintenance Ticket.”
3. System requests room number / geo-tag, issue category, severity, and photo attachment.
4. Resident enters the required details and attaches the photo.
5. System validates the information.
6. System creates the ticket and generates a unique ticket ID.
7. System routes the ticket to the Maintenance Warden queue.
8. System displays confirmation with the ticket ID and current status.
9. Use case ends successfully.

ALTERNATE FLOW :

- Missing room tag: The system rejects submission and asks the resident to provide the room/location tag.

. Missing/invalid required information: The system highlights the field and prevents ticket creation until corrected.

. Ticket creation failure: The system displays an error and asks the resident to retry; no incomplete ticket is routed.


Hostel maintenance and issue ticketing system
Login
«include»
Raise maintenance issue
«include»
view issue status
«include»
view / manage status
«include»
Assign maintenance staff
«include»
update issue status
«extend»
Escalate Overdue ticket
Generate Maintenance Report
Hostel Resident
maintenance warden
Hostel admin